

1. (b) Data and a pointer to the next node.
2. (c) NULL
3. (b) Self-referential structure
4. (b) head == NULL
5. (b) newnode->next = head; head = newnode;
6. (b) temp = head; head = head->next; free(temp);
7. (c) Traversing from tail to head
8. (b) n
9. (c) copying the data of p->next into p, then deleting p->next
10. (b) while (p != NULL)
11. (b) p->next == NULL
12. (c) two pointers advancing at different rates
13. (c) Three
14. (b) Exactly one node
15. (b) p = (struct node*) malloc (sizeof (struct node));
16. (b) Avoid special case handling for insertion and deletion at the head.
17. (c) The first node
- 18.
19. (b) loss of access to all nodes, causing a memory leak.
20. (c) A slow pointer and a fast pointer.
21. (c) The last node.

```
22. #include <iostream>
    struct node {
        int data;
        struct node* next;
    };

```

24. Output : 30 50 70

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25. struct node* reverselist(struct node* head){

struct node * prev = NULL;

struct node * curr = head;

struct node * next = NULL;

while (curr != NULL) {

next = curr -> next;

curr -> next = prev;

prev = curr;

curr = next;

}

return prev;

}

28. 5 → 8 → 12 → 3 → NULL

7 → 5 → 8 → 12 → 3 → NULL

7 → 5 → 8 → 3 → 9 → NULL

Final → 7 → 5 → 8 → 3 → 9 → NULL